Create a list of the first 10 positive integers and print the sum of the even numbers in the list

```
num=list(range(1,11))

list

sum=0
for i in num:
    if(i%2==0):
        sum+=i
print(sum)

30
```

Demonstrate the following operations on a number list using list functions a.insert 13 at position 4 b.Sort the list in ascending order c.delete the last element d.remove 31 e.Reverse the list f.Append one number to list g.extend the list with [20,30,40] h.print the number of elements in the list using while

```
num=list(range(1,11))
print(num)

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

num.insert(4,13)
print(num)

[1, 2, 3, 4, 13, 5, 6, 7, 8, 9, 10]

num.sort()
print(num)

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 13]

num.pop()
print(num)

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

num.remove(31)

---------------------------------------------------------------------------
-----
ValueError                                Traceback (most recent call
last)
<ipython-input-25-04da6c47085e> in <module>
----> 1 num.remove(31)

ValueError: list.remove(x): x not in list
```

```
num.sort(reverse=True)
print(num)
```

```
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

```
num.append(45)
print(num)
```

```
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 45]
```

```
l=[20,30,40]
num.extend(l)
print(num)
```

```
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 45, 20, 30, 40]
```

```
i=0
while i < len(num):
    print(num[i])
    i+=1
```

```
10
9
8
7
6
5
4
3
2
1
45
20
30
40
```

Given the tuple (10, 20, 30, 40, 50), use slicing to print a new tuple containing only the last three elements.

```
num=(10,20,30,40,50)
print(num[2:])
```

```
(30, 40, 50)
```

Create a set of the numbers {1, 2, 3, 4, 5} and compute the intersection with {4, 5, 6, 7}; print the result.

```
set1={1,2,3,4,5}
set2={4,5,6,7}
```

```
print(set1&set2)
```

```
{4, 5}
```

Define a dictionary with the keys 'a', 'b', and 'c' and their respective values 1, 2, and 3, then update the value for key 'b' to 10 and print the updated dictionary.

```
thisdict ={
    'a' : 1,
    'b' : 2,
    'c' : 3
}
print(thisdict)
dict['b']=10
print(thisdict)
```

```
{'a': 1, 'b': 2, 'c': 3}
{'a': 1, 'b': 2, 'c': 3}
```

Using a list comprehension, create a list of squares for the numbers from 1 to 5 and print the resulting list.

```
sq = [x**2 for x in range(1,6)]
print(sq)
```

```
[1, 4, 9, 16, 25]
```

Create a tuple with elements 'cat', 'dog', and 'bird', then check and print whether 'dog' is in the tuple.

```
animal=("cat","dog","bird")
print("dog" in animal)
```

```
True
```

Given the list [1,1,2,3,5,8,13], convert it to a set and print the set's length.

```
ls = [1,1,2,3,5,8,13]
print(ls)
s = set(ls)
print(s)
print(len(s))
```

```
[1, 1, 2, 3, 5, 8, 13]
{1, 2, 3, 5, 8, 13}
6
```

Create two dictionaries {'A': [1, 2], 'B': [3, 4]} and {'B': [5, 6], 'C': [7, 8]}, then merge them into one dictionary and print the result.

```
dict1 = {'A':[1,2], 'B':[3,4]}
dict2 = {'B':[5,6], 'C':[7,8]}
merged_dict = {**dict1, **dict2}
print(merged_dict)
```

```
{'A': [1, 2], 'B': [5, 6], 'C': [7, 8]}
```

For the list [3, 5, 7, 3, 5, 9, 3], create a dictionary to count the occurrences of each element and print the dictionary.

```
elements = [3, 5, 7, 3, 5, 9, 3]
counts = {}

for num in elements:
    if num in counts:
        counts[num] += 1
    else:
        counts[num]=1
print(counts)
```

```
{3: 3, 5: 2, 7: 1, 9: 1}
```